

Can Pro-Crypto Policy Cause Remittance Prices and Transfer Speed to Drop?
Analyzing the Effect of Lifting a Crypto Ban in Nigeria on Remittance Prices and Speed

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Literature Review

In 2023 the remittance market was \$857 billion (World Bank, 2024). The global average cost of sending \$200 across borders in Q1 of 2025 was 6.49% (World Bank, 2025). In 2025 Q1 it took, on average, 2.39 days to send money internationally (World Bank, 2025). These figures represent the importance of remittances yet the lack of efficiency in the market. Freund and Spatafora (2008) explain that remittance prices remain structurally high due to limited competition among service providers, exclusive partnerships between banks and money transfer operators (MTOs), and regulatory frictions related to anti-money-laundering compliance. Beck and Martinez Peria (2011) similarly find that market structure, particularly bank dominance, plays a significant role in driving remittance prices upward. Cryptocurrencies offer a solution to this costly problem by allowing anyone with internet access to send money anywhere with minimal fees and almost instantaneously. Right now stablecoins especially are becoming more popular, especially with the US government passing the GENIUS Act in July 2025. As digital currencies grow and gain government support, this could significantly decrease the price and speed it takes to send money across borders, leading to countries who rely on remittances to grow their economies and increase the wealth of remittance dependent individuals.

High remittance pricing and slow processing speed disproportionately affects developing countries, many of which rely heavily on remittance inflows to support their economy. Tajikistan was the most remittance dependent country in 2024 where their GDP was 45.4% remittances (IMF, 2024). Despite the economic importance of remittances, empirical evidence historically suggests the market remains inefficient, with slow processing times and costly intermediaries (Agarwal & Horowitz, 2002). These inefficiencies create natural policy motivation for exploring

new technologies, such as digital currencies and stablecoins, that could lower transaction costs and accelerate settlement.

A rapidly growing literature examines the potential role of cryptocurrencies, stablecoins, and blockchain based systems in lowering remittance frictions. Advocates argue that crypto networks, particularly stablecoin rails, can enable relatively cheap and fast transfers (Narula, 2021). The Bank for International Settlements (BIS) finds that blockchain based cross border payments can reduce counterparty and settlement risks while lowering compliance costs through programmable verification (Auer et al., 2022).

Empirical evidence remains mixed, however. Mohan and Narayan (2022) note that while crypto platforms theoretically enable cheaper remittances, adoption barriers, including regulatory uncertainty, volatility, and limited financial literacy, constrain widespread substitution away from traditional channels. In countries with restrictive crypto policies, remittance senders face additional transaction risks associated with converting between fiat and digital assets. Case studies also illustrate the heterogeneity in crypto remittance experiences. El Salvador's 2021 Bitcoin adoption produced mixed results: despite significant government promotion, Bitcoin usage for remittances remained low due to volatility, poor user interface design, and inconsistent internet access (Alvarez et al., 2022). In contrast, the Philippines and parts of Latin America have experienced an increase in remittances via crypto, facilitated primarily through fintech platforms rather than national governments (Chainalysis, 2023).

Overall, the literature points to two consistent channels through which digital currency policy could affect remittance outcomes. The first being transaction costs where blockchain networks can reduce the number of intermediaries needed for cross border payments. Secondly, the transfer speed can be significantly reduced, compared to days for SWIFT based transfers.

Finally, legalizing and supporting digital currencies may pressure traditional MTOs and banks to reduce their fees. These mechanisms provide the theoretical foundation for my hypothesis that Nigeria's decision to lift its crypto ban could reduce the price and speed of receiving remittances.

Nigeria

Nigeria has been the subject of extensive economic research due to its chronic foreign exchange shortages, volatility, and regulatory experimentation. Prior to December 2023, Nigeria operated under a complex system of multiple exchange rates, which produced substantial arbitrage opportunities and reduced confidence in the formal banking sector (IMF, 2023). Research shows that foreign exchange restrictions often drive remittance senders toward informal or black market channels, increasing both cost and risk (Saxegaard, 2006).

The Central Bank of Nigeria (CBN) has historically taken a restrictive stance toward cryptocurrencies, issuing a 2017 directive prohibiting banks from interacting with digital asset companies and reinforcing that ban in 2021. Studies of Nigeria's fintech sector, however, show that crypto adoption continued informally, driven by inflation, currency instability, and a pervasive need for alternative savings instruments (Ndukwe, 2021). Nigeria's lift of the crypto ban in December 2023 fits within a broader policy effort to liberalize the exchange rate and address macroeconomic instability. High inflation, above 20 percent in 2024, and limited financial inclusion (with over one third of adults unbanked) make Nigeria an interesting case for the effect of pro-crypto policy on remittances (CBN, 2024). Ambrosius and Widmaier's research on financial development suggests that improving access to mobile and digital financial services can reduce remittance costs and improve efficiency (2020).

Egypt

Egypt shares many macroeconomic characteristics with Nigeria, strengthening its role as a natural control group. The Central Bank of Egypt maintained a managed float foreign exchange (FX) regime until March 2024, when it adopted a fully market driven exchange rate as part of an IMF stabilization program (IMF, 2024). Literature on Egypt's FX interventions highlights persistent currency pressures, remittance dependence, and the importance of exchange rate reform for restoring macroeconomic balance (IMF, 2024). Like Nigeria, Egypt has faced chronic inflation, averaging above 10 percent in recent years (CNE, 2025). However, unlike Nigeria, Egypt has not altered its cryptocurrency restrictions since 2017. The absence of major crypto policy changes makes Egypt suitable as a control group for evaluating Nigeria's ban lift.

I will use a difference-in-differences (DID) model with Nigeria as the treatment group and Egypt as my control, the shock being the fourth quarter of 2023 when Nigeria lifted its crypto ban. DID validity relies on the parallel trends assumption, that treated and control units would have followed similar trends absent the intervention (Angrist & Pischke, 2009). However, a key methodological challenge in DID with country level treatments is the small number of clusters. Cameron and Miller (2015) show that when treatment occurs at the aggregate level, standard cluster robust standard errors often severely understate uncertainty. MacKinnon and Webb (2017) propose wild cluster bootstrap methods as a solution, demonstrating that Rademacher weights can produce degenerate results when the number of clusters is extremely small, particularly when $G \leq 3$.

My DID only includes two clusters, one for Nigeria and one for Egypt. As expected, Rademacher bootstrap p-values became artificially small because there are only four possible bootstrap draws (2^2). Webb weights, which use a six point distribution, perform better under

small cluster conditions and generate more conservative and realistic inference. However, the literature makes clear that no statistical technique can fundamentally overcome the inferential limitations of two cluster DID designs (Abadie et al., 2023). Thus, results must be interpreted cautiously.

My project contributes to the established literature by: (1) evaluating whether pro-crypto policy affects transaction cost or speed, this study adds empirical evidence on mechanisms to reduce global remittance frictions and (2) examining the effects of lifting a ban rather than adopting a cryptocurrency as legal tender (like in El Salvador), offering a different perspective on regulatory liberalization. By examining Nigeria's policy through a credible comparison with Egypt, this study provides evidence on how encouraging the usage of digital currencies in formal financial channels interacts with remittance markets in developing economies.

Question: What is the effect of Nigeria lifting its crypto ban in 2023 on the cost and speed of receiving remittances in Nigeria, relative to Egypt?

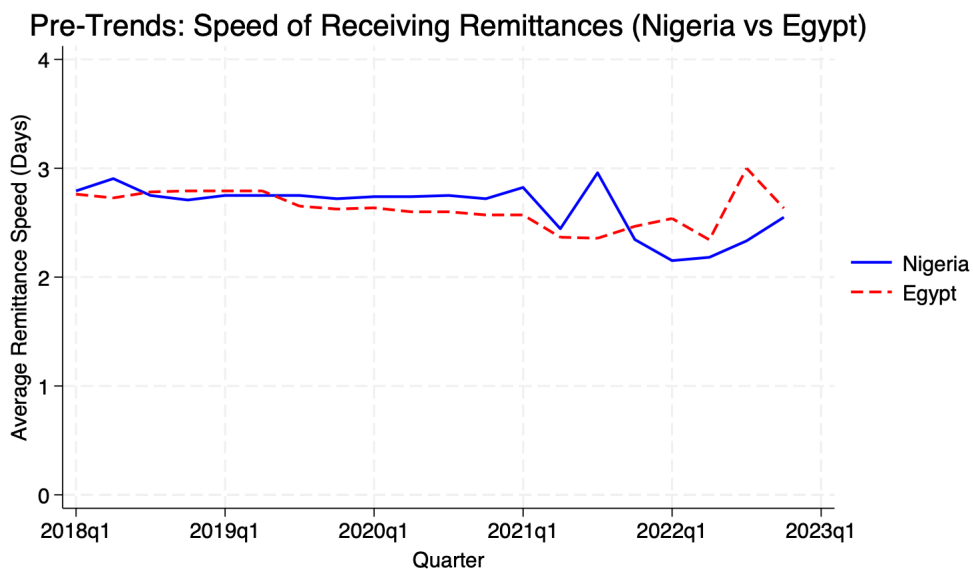
Data

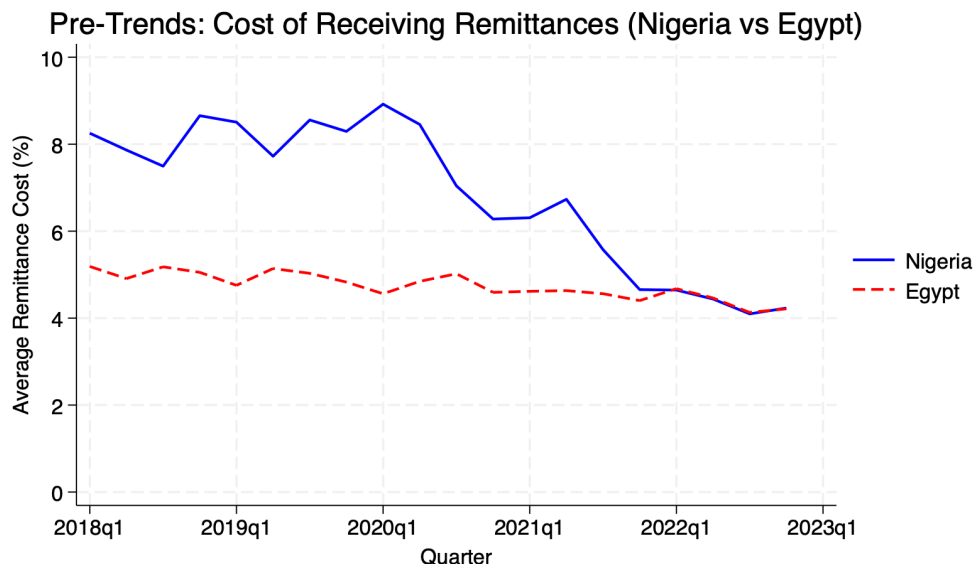
My dataset is from the World Bank and contains 247,490 observations from 2021 Quarter 1 to 2025 Quarter 1 with 115 different countries receiving remittances. The variables I include in my analysis are the receiving country's name, the cost of receiving money as a percentage, the time it takes to receive money in days (hours converted to days), and time period in quarterly units. Specifically, I am looking at Nigeria as my treatment group and Egypt as my control group. The dataset includes 6,110 observations for Nigeria and 4,157 observations for Egypt. Below are the summary statistics for my treatment and control groups.

Nigeria - Treatment	Mean	St. Deviation	Min	Max
Cost (%)	5.673633	1.483888	4.097619	8.920382
Speed (Days)	2.55613	0.2729796	2.148148	2.958333

Egypt - Control	Mean	St. Deviation	Min	Max
Cost (%)	4.485569	0.258073	4.130606	5.020417
Speed (Days)	2.561937	0.1554704	2.342857	3

The graphs below illustrate the preshock trends for Nigeria and Egypt. The speed trends are very similar up until 2021 where they start to fluctuate a bit, however both trends stay on average between 2-3 days. The opposite is displayed on the preshock trends for cost. Nigeria and Egypt start far apart with Nigeria being significantly more costly to send money to, until the last quarter in 2021 where they converge to almost exactly the same line up until 2023 when the shock occurs.





The comparability between Nigeria and Egypt strengthens the credibility of the difference-in-differences design and supports the parallel trends assumption. Both countries rely heavily on remittances as a source of household income and foreign exchange, and both entered the study period facing severe macroeconomic imbalances: high inflation, currency pressures, and structural economic instability. Importantly, while Nigeria implemented a major policy shift by lifting its cryptocurrency ban, Egypt did not enact similar digital asset reforms, making it unlikely that Egypt's remittance prices or processing times were affected by Nigeria's policy change. At the same time, both economies undertook exchange-rate liberalization efforts around the same general period, meaning that broader regional or global shocks would plausibly affect the two countries in similar ways. These similarities in economic structure, vulnerability to external shocks, and institutional responses create a setting in which Nigeria and Egypt are expected to follow comparable pre-treatment trends in remittance costs and speeds. This increases confidence that any divergence observed after Nigeria's policy reform reflects the causal impact of the ban lift rather than unrelated country-specific developments.

Empirical Strategy

Using a difference-in-differences approach I ran the following two way fixed effect regression:

$$Y_{ct} = \alpha_c + \lambda_t + T(\text{Treated}_c * \text{Post}_t) + \varepsilon_{ct}$$

Here, time effects and country specific effects are controlled by λ_t and α_c , respectively. The treatment effect is T , representing the effect of Nigeria lifting its crypto ban on the cost and speed of receiving remittances in Nigeria. Treated_c is a dummy variable equal to 1 if the country is Nigeria, Post_t is another dummy variable equal to 1 if the time period is quarter 4 of 2023 or later. In order to interpret these results causally, I am assuming (1) that parallel trends hold, (can reference my pre-treatment trends graph above). This model also assumes (2) that there was no anticipation in Nigeria of the crypto ban being lifted. The third assumption is (3) that the ban lifted did not result in any spillovers to Egypt. Remittances in Egypt were not affected by the ban lift in Nigeria, because Egypt still is restricting crypto. I also assume (4) that the shock was exogenous. This assumption holds because Nigeria lifted their previous crypto ban to combat high inflation and depreciation rates. Finally, (5) the model assumes fixed effects controls for all country specific shocks and global time shocks (Angrist and Pischke, 2009). The results should still be interpreted cautiously due to the inherent limitations of having only two clusters. The results are described in the table below.

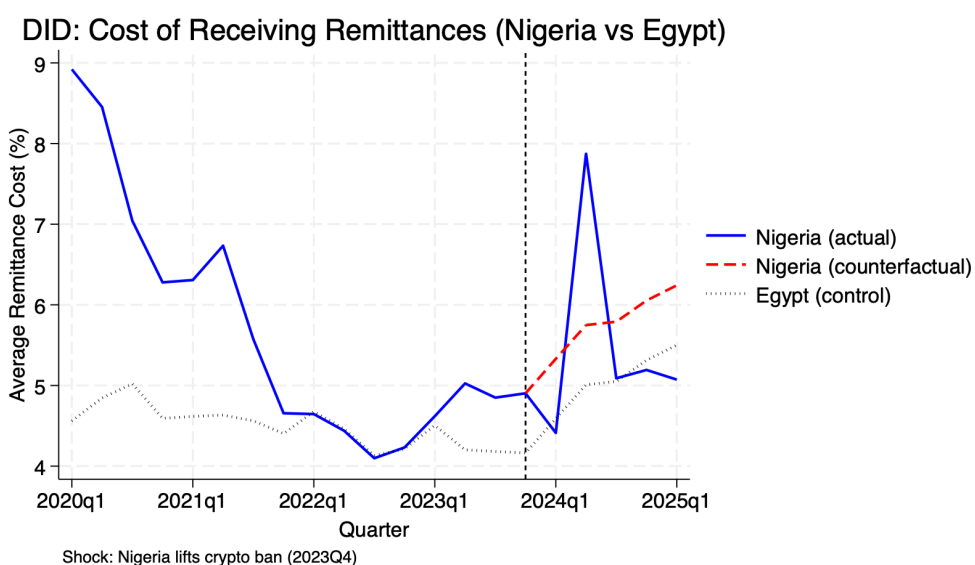
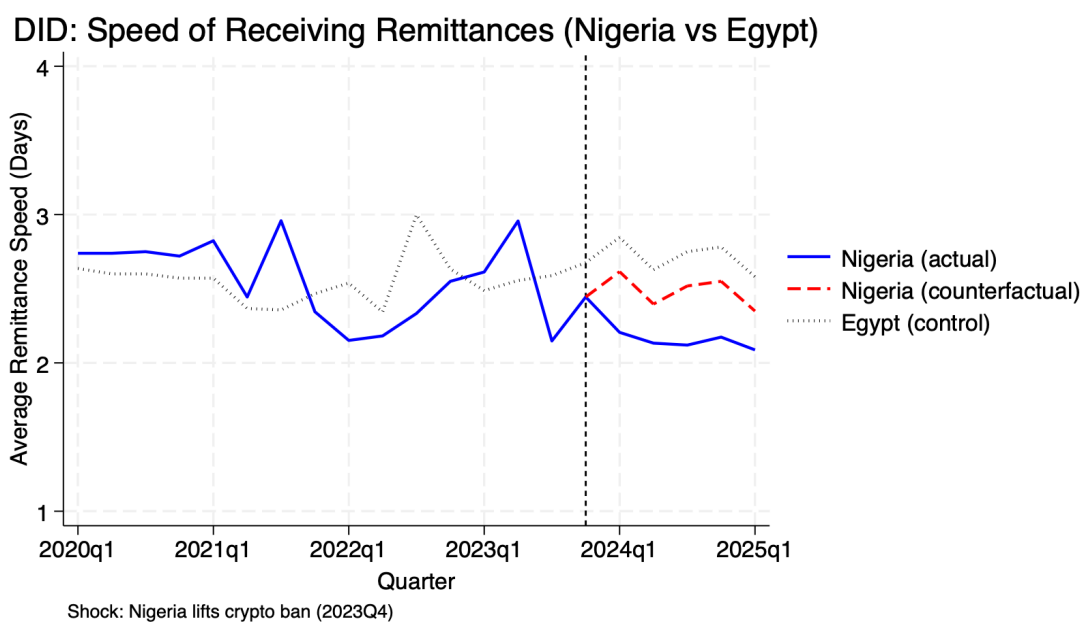
For Speed:

Treatment Effect	Rademacher p-value	Webb p-value
-1.051	0.0000	0.22

For Cost:

Treatment Effect	Rademacher p-value	Webb p-value
-2.47	0.0000	0.27

This table shows that, under the assumptions stated above, the 2023 shock in Nigeria caused the time it took for Nigerians to receive remittances by approximately 1 day and the cost to decrease by 2.47 percentage points, relative to Egypt. However, neither result is statistically significant so we cannot reject the null hypothesis that the effect is zero. So inference is weak, due to the fact that there are only two clusters in the regression model. These effects are also displayed in the difference-in-differences plots below.



Interpretation

The speed of receiving remittances in Nigeria falls consistently after the shock. On the contrary, the cost of receiving remittances in Nigeria shoots back up in the second quarter of 2024, but does initially drop right after the shock. This could be due to a mandate that the CBN passed on March 28, 2024 raising the minimum capital requirement for banks in an effort to strengthen their financial system (CBN, 2024). This increase in the minimum capital requirement may have caused banks in the short run to raise their fees in order to quickly increase their capital.

Pro-crypto policy is a potential mechanism for lowering remittance rates and speed which could significantly increase the wealth for high impact countries. However, by only using Egypt as a single control against Nigeria, I am left with only two clusters in my regression analysis. This leads to unreliable p-values and therefore a weak inference. For the next stages of my project, I plan to look into using synthetic control by using all countries that still restrict crypto usage.

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